

# PROJECT DESIGN PHASE

## Proposed Solution Template

|                     |                        |                      |         |
|---------------------|------------------------|----------------------|---------|
| <b>Date</b>         | 12 June 2026           | <b>Team ID</b>       |         |
| <b>Project Name</b> | E-commerce Application | <b>Maximum Marks</b> | 2 Marks |

*Proposed Solution Template: Project team shall fill the following information in the proposed solution template.*

| S.No. | Parameter                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>Problem Statement<br/>(Problem to be solved)</b> | Traditional physical retail and fragmented online storefronts struggle to provide an integrated, responsive shopping experience, resulting in high checkout friction and complex inventory discrepancies. Small-to-medium businesses lack an all-in-one scalable ecosystem to efficiently track multi-channel inventory, securely handle real-time payment transactions, and derive actionable sales insights. |
| 2     | <b>Idea / Solution<br/>description</b>              | ShopEZ is a robust e-commerce platform offering an end-to-end purchasing environment integrated with dynamic catalog filtering, customer feedback loops, and highly responsive shopping carts. The backend architecture features automated inventory reconciliation tools, secure multi-tiered checkouts, and a unified administrative dashboard providing live analytical charts on sales patterns.           |
| 3     | <b>Novelty / Uniqueness</b>                         | Unlike monolithic platforms, ShopEZ implements an optimized caching layer for seamless high-speed catalog browsing alongside a modular predictive inventory alert module. This structure bridges the gap between client usability and warehouse execution, maintaining real-time transactional accuracy even under synchronous multi-user stress conditions.                                                   |
| 4     | <b>Social Impact /<br/>Customer Satisfaction</b>    | The application empowers regional merchants by granting them immediate access to digital market spaces, expanding localized commerce and generating sustainable livelihoods. Customers benefit from intuitive, localized accessible user interfaces, fully encrypted secure checkouts, and reliable review transparency that actively prevents fraudulent listings.                                            |
| 5     | <b>Business Model<br/>(Revenue Model)</b>           | The platform operates on a multi-tiered monetization strategy consisting of a basic commission per successful purchase transaction from onboarded corporate vendors. Additional value-driven streams                                                                                                                                                                                                           |

| S.No. | Parameter                          | Description                                                                                                                                                                                                                                                                                                                                                    |
|-------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |                                    | include tiered premium cloud subscription tiers for advanced storefront customizations and localized target-market advertisement placements.                                                                                                                                                                                                                   |
| 6     | <b>Scalability of the Solution</b> | Designed using a modular, decoupled software framework, the system horizontally scales its catalog and transactional nodes to accommodate sudden traffic surges during seasonal events. Database micro-sharding and asynchronous queue handling ensure that analytical parsing remains distinct from transactional workflows, preventing server bottlenecking. |